

Bingqi Shang

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RESEARCH INTERESTS

Trustworthy Machine Learning: Machine Unlearning, Alignment & RLHF, Adversarial Machine Learning

Generative AI: Large-language Model, Multi-modal Language Model, Reasoning, Post-training, On-policy Distillation

EDUCATION

Michigan State University (MSU)

Ph.D., Computer Science

Sep. 2025 - Present

Advisor: [Prof. Sijia Liu](#)

Northwestern University (NU)

M.S., Computer Science

Sep. 2023 - Jun. 2025

Advisors: [Prof. Qi Zhu](#) and [Prof. Xiao Wang](#)

Tongji University

B.E., Software Engineering

Sep. 2019 - Jun. 2023

[School of Computer Science and Technology](#)

RESEARCH EXPERIENCE

LLM Lineage Tracing via Spectral Fingerprints

Nov. 2025 - Mar. 2026

Supervisor: [Prof. Sijia Liu](#) (MSU)

- Proposed a data-free framework to trace LLM lineage from **weight geometry** alone, without access to training data or model inputs.
- Showed that spectral energy separates independently trained model families, while subspace alignment discriminates models built on a shared base, across **110+** open-weight model pairs.
- **Publication:** [1].

Backdooring Machine Unlearning via Attention Sinks

Apr. 2025 - Oct. 2025

Supervisor: [Prof. Sijia Liu](#) (MSU)

- Exposed a vulnerability in LLM unlearning: attention-sink-guided backdoors can **reactivate** forgotten behaviors.
- Exploring applications in unlearned models where backdoor triggers can selectively recover forgotten knowledge.
- **Publication:** [2].

Privacy-Preserving Tuning for Large Models

Dec. 2023 - Mar. 2025

Supervisors: [Prof. Qi Zhu](#) (NU), [Prof. Xiao Wang](#) (NU)

- Developed Split Adaptation (SA) to ensure **data privacy** during adaptation of pre-trained Vision Transformers (ViTs), utilizing bi-level noise injection for privacy-preserving downstream tasks without data sharing.
- Protected **model privacy** by sharing only a low-bit quantized frontend of the ViT, preventing model leakage and ensuring secure adaptation.
- **Publication:** [3].

PROFESSIONAL EXPERIENCE

MIT-IBM Watson AI Lab *Cambridge, MA*

Jun. 2026 - Present

Research Scientist Intern, Mentor: [Yang Zhang](#)

SAP *Shanghai, China*

Jun. 2022 - Mar. 2023

Cloud Developer Intern, Supervisor: April Qi

Project: Cloud Provider Exporter in Go on Kubernetes for AWS, Azure, and GCP, using Prometheus and Grafana.

PUBLICATIONS

* indicates equal contribution

[1] Yiwei Chen*, [Bingqi Shang*](#), Sijia Liu. [Who Built This Model? Tracing LLM Lineage via Spectral Fingerprints in Weight Space](#). *COLM'2026*.

[2] [Bingqi Shang*](#), Yiwei Chen*, Yihua Zhang, Bingquan Shen, Sijia Liu. [Forgetting to Forget: Attention Sink as A Gateway for Backdooring LLM Unlearning](#). *COLM'2026*.

- [3] Lixu Wang*, **Bingqi Shang***, Yi Li, Payal Mohapatra, Wei Dong, Xiao Wang, Qi Zhu. **Split Adaptation for Pre-trained Vision Transformers**. *CVPR'2025*.
- [4] Yancheng Huang*, Changsheng Wang*, Chongyu Fan, Yicheng Lang, **Bingqi Shang**, Yang Zhang, Mingyi Hong, Qing Qu, Alvaro Velasquez, Sijia Liu. **Subspace Control: Turning Constrained Model Steering into Controllable Spectral Optimization**. *COLM'2026*.
- [5] Zijie Liu, Jinhao Duan, **Bingqi Shang**, Xinming An, Sijia Liu, Tianlong Chen. **When Retain Constraints Conflict: Mitigating Forget-Retain Interference in Tabular Data**. *EMNLP'2026 Findings*.

HONORS

- Shanghai Outstanding Graduate Award 2023
- Outstanding Undergraduate Dissertation Award of Tongji University 2023
- **National Scholarship** (Top 0.2%, highest undergraduate honor in China) 2020

SERVICES

Workshop Co-organizer: AdvML-Frontiers × CoTMA @ COLM'26

Journal Reviewer: IEEE TSP, IEEE TAES

Conference Reviewer: NeurIPS 2026, CVPR 2026, ICASSP 2026–2027

PERSONAL INTERESTS

Photography 2019 - Present

PROFESSIONAL SKILLS

Programming Languages: Python, Go, C++, Java, Rust, L^AT_EX, HTML, CSS, JavaScript

Machine Learning Systems: PyTorch, Transformers, Huggingface, W&B, OpenCV, Scikit-learn